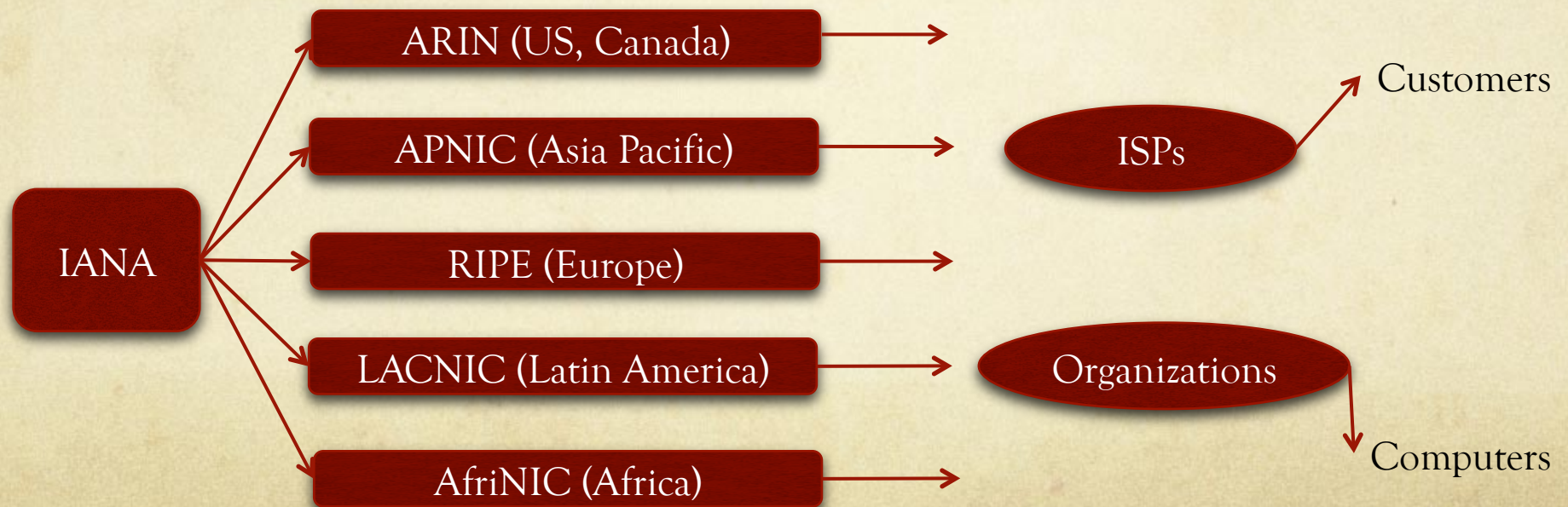


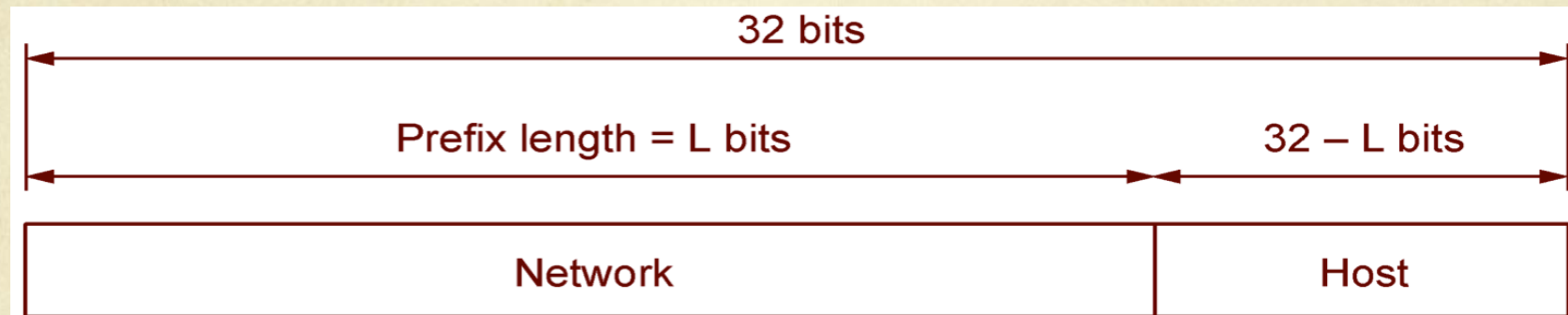
IP address

- ◆ Public IP addresses are *allocated* & *managed* hierarchically
 - ◆ *IANA* delegates blocks of addresses to *regional authorities* (*RIR*)
 - ◆ *ISPs/organizations* get blocks of addresses from RIRs
 - ◆ ISPs/organizations assign addresses to *customers/computers*



IP address

- ◆ IP addresses are allocated in contiguous blocks called *prefixes*



- ◆ Most significant L bits form *prefix* or *network* address
- ◆ $(32 - L)$ bits identify *host* within network or *subnet*
 - There are $2^{(32-L)} - 1$ hosts in subnet

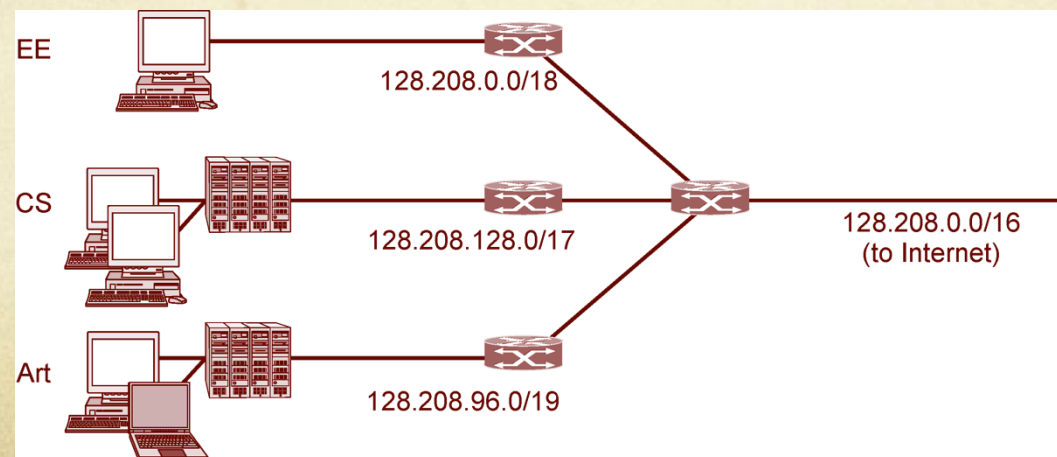
IP address

- ◆ *Prefix* representation
 - ◆ /prefix length in bits → E.g., */24* for 24-bit prefix

- ◆ *Network* of computers can be represented using
 - ◆ Start address within network/length of prefix in bits
 - ◆ E.g., *172.16.0.0/24*
 - ◆ Least significant *8 bits* form host identifier
 - ◆ Address range is *172.16.0.0* to *172.16.0.255*

Subnetting

- ◆ Organizations can partition their own block of addresses
 - ◆ Create smaller *subnets* within their *assigned* subnet
- ◆ E.g., consider a university network **128.208.0.0/16**
 - ◆ EE.: 10000000 11010000 **00** | xxxxxxx xxxxxxxx
 - ◆ CS: 10000000 11010000 **1** | xxxxxxxx xxxxxxxx
 - ◆ Art: 10000000 11010000 **011** | xxxxxx xxxxxxxx



Subnetting

- ◆ *Shorter* prefix
 - ◆ Larger block of IP addresses with same prefix
 - ◆ *Less specific* prefix

- ◆ *Longer* prefix
 - ◆ Smaller block of IP addresses with same prefix
 - ◆ *More specific* prefix